

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming With Java

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Abstract

The Social Media Feed System is a console-based Java application developed using Core Java and Object-Oriented Programming (OOP) concepts. The system allows users to interact with posts by liking and commenting on them. It displays all posts along with their likes and comments in a feed format. The project helps in understanding how social media platforms manage user interactions and content. It also demonstrates important OOP concepts such as classes, objects, encapsulation, methods, and ArrayList.

Introduction

Social media platforms allow users to share content and interact with others through likes and comments. The Social Media Feed System is a simple Java application that simulates these basic features. Users can like posts, add comments, and view all posts in a feed. The project is designed to provide practical experience with OOP concepts and class interactions in Java.

Features

- Create and manage posts
- Like posts
- Add comments to posts
- Display posts in feed format
- Show total likes and comments

Objectives

- Manage social media posts efficiently
- Allow users to like and comment on posts
- Display posts in a structured feed
- Apply OOP concepts in Java
- Use ArrayList for storing comments dynamically

Scope

This project is suitable for students learning Java programming and OOP concepts. It can be extended in the future by adding database support, login functionality, and graphical user interfaces.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- VS Code, Eclipse, or IntelliJ IDEA

Methodology

1. Create User and Post objects.
2. Users interact with posts through likes and comments.
3. Comments are stored using ArrayList.
4. Like count is updated whenever a post is liked.

5. All posts are displayed in a feed showing content, likes, and comments.

Implementation

User Class: Stores user details and provides methods for liking and commenting on posts.

Post Class: Stores post content, total likes, and comments. It includes methods to add likes, add comments, and display post information.

SocialMediaApp Class: Contains the main method and manages the overall flow of the application.

Future Enhancements

- User login and registration
- Edit and delete posts
- Database integration using MySQL
- GUI using Java Swing or JavaFX
- Track users who liked posts

Conclusion

The Social Media Feed System successfully demonstrates the implementation of Core Java and OOP concepts. It provides a simple platform where users can interact with posts through likes and comments. The project improves understanding of object interaction, encapsulation, and dynamic data handling in Java.

References

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- [3] Oracle, Java Documentation. Available: <https://docs.oracle.com>